

Point-by-point response to R2

The reviewer’s original comment is reproduced under **Reviewer comment**. Our response, new evidence, and revision action follow under separate labels so that the editorial record is auditable.

R2-1

Reviewer comment The full estimator pipeline (specificity weighting, platform self-gate, Poisson closure; Section 3, lines 118-135) is inherited entirely from a single self-cited reference, “Fu, 2026,” described as a Zenodo software archive dated in the future relative to this submission. Since the present manuscript’s contribution is explicitly framed as two additions to that stack, its own validity is contingent on a method that appears not to be independently peer-reviewed. Please (a) clarify the publication/review status of Fu (2026), and (b) briefly summarize in this manuscript how the base estimator was itself validated, so the present work is interpretable as a self-contained contribution.

Authors’ response and evidence

Comment. The reviewer describes Fu (2026) as a separate unpublished baseline. We have checked the repository identity and the code boundary and correct that framing.

Response. SPGD-TME is the standalone Methods contribution and public repository under review. Its scientific contribution is the two locked operators, their decision contract, and the cross-library evidence: (i) the malignant-neighbor KEEP/ABSTAIN reportability gate and (ii) the Patient_ID-disjoint donor split. Fu (2026), the GitHub/Zenodo software record cited in the manuscript, is the released SPGD baseline implementation used as a fixed computational stage within this same software lineage; it is not an invisible external authority to which the present contribution outsources its validity. The revised Methods now states the baseline equations and constants needed to interpret the two operators, explicitly labels baseline-stage provenance separately from operator-inclusive evidence, and the Results and new Figures 13–14 provide evidence generated by this manuscript. We therefore retain the provenance citation but remove the former dependency language from the V2 manuscript and portal response.

The software boundary is stated precisely. The public SPGD-TME repository is independently archived and version-checked, while its wrapper currently imports the released SPGD baseline implementation from the sibling public SPGD software archive. That import boundary affects raw-data-to-fit packaging; it does not change the scientific identity of the two operators evaluated here. We do not claim that the repository vendors every baseline dependency, and we do not claim that the baseline is a new contribution of this article.

New evidence. The repository audit confirms the public source repository, the released baseline record, and the fixed-baseline import boundary. The manuscript now exposes the baseline constants and definitions needed to interpret the two operators, while Figures 13–14 report evidence generated by this manuscript. This is a provenance clarification, not a transfer of the scientific contribution to another archive.

Revision action. Replace the former dependency wording with “standalone SPGD-TME Methods contribution” plus “released SPGD baseline implementation”; retain Fu (2026) as software provenance; add the explicit three-layer distinction (scientific contribution, independent archive, fixed-baseline provenance) to Methods and the response. The reviewer’s concern is therefore answered as an identity/provenance clarification, not as a concession that the paper is merely an extension of a separate unpublished paper.

R2-2

Reviewer comment The manuscript states repeatedly (e.g., lines 47-48, 165, 330) that $c^* = 0.80$ is “frozen before any sweep,” which is good practice against post hoc threshold selection. However, no rationale is given anywhere in the manuscript for why 0.80 specifically was chosen. Given that this threshold is the paper’s central mechanistic claim, please add: (a) the origin/derivation of 0.80, and (b) a sensitivity analysis showing whether the eight KEEP/ABSTAIN calls are stable for c^* in a plausible neighborhood (e.g., 0.75-0.85).

Authors’ response and evidence

Comment. Why $c^*=0.80$, and whether calls are stable in 0.75–0.85.

Response. We agree that freezing a threshold protects against post hoc tuning but does not, by itself, establish why 0.80 was selected. We therefore retain 0.80 as the historical locked operating point and explicitly state that this study did not retrospectively optimize it by ROC analysis. We evaluated the prespecified 0.75–0.85 neighborhood and separate the answer into two estimands: the eight historical designated library calls are stable throughout the neighborhood, whereas complete-neighbor maxima are not. At the locked cutoff, the reconstructed maxima are 0.847 for NSCLC, 0.791 for CRC, 0.854 for HCC, and 0.905 for PDAC. This sensitivity result motivates transparent reporting of the rule dependence; it is not used to retune the cutoff.

New evidence. All eight historical designated calls remain unchanged across 0.75–0.85. At $c^*=0.80$, complete-scan maxima are NSCLC 0.847336, CRC 0.791305, HCC 0.854095, PDAC 0.905088; a max rule would change NSCLC/HCC.

Revision action. Methods cutoff provenance paragraph; Results sensitivity paragraph; Supplementary Table S3 with threshold grid and rule comparison. Evidence location: Fig. 13A–D and the complete threshold/neighbor audit table.

R2-3

Reviewer comment For each library, a single designated neighbor pair is scored. Figure 12B/F indicates that all annotated types were in fact scanned per library, with a variable fraction exceeding the cutoff. Is the reported cosine defined algorithmically as the maximum over all annotated non-malignant types, or is it a pair pre-selected by the authors on biological grounds? If the latter, what rule determined that selection, and is it applied identically before results are seen? Please state this selection rule explicitly and formally as an equation or algorithm step.

Authors’ response and evidence

Comment. How is the malignant-neighbor pair selected?

Response. We agree that this is the most important reproducibility issue in the current implementation. The historical scripts do not implement one uniform selection rule: they contain preference-ranked neighbors, a first-KEEP decision, an HCC-specific preference mutation, and a CRC export with incomplete coverage. We therefore do not silently rewrite the historical calls. We have now constructed a candidate author-lock rule: define the eligible set before looking at outcomes, score every eligible non-malignant program, use the largest cosine at the frozen cutoff, and resolve exact ties deterministically. The complete available scan shows that this candidate rule changes NSCLC and HCC relative to the historical designated pair, while CRC remains KEEP and PDAC remains ABSTAIN. The review package now displays historical and

candidate-rule analyses separately; corrected downstream fits are deliberately withheld until this operating rule is formally locked, so no candidate outcome is presented as the primary result.

New evidence. The candidate rule table now contains 15 eligible CRC pairs, 17 NSCLC pairs, 14 HCC pairs, and 5 PDAC pairs. At $c^*=0.80$, the candidate maxima are BEC 0.791 (CRC, KEEP), epithelial 0.847 (NSCLC, ABSTAIN), CD3+ alpha/beta T cells 0.854 (HCC, ABSTAIN), and macrophage 0.905 (PDAC, ABSTAIN). BCC, openST, Xenium, and Xenium FLEX remain designated-only until their complete eligible scans are available. These are selection-audit results; no corrected downstream composition is claimed yet.

Revision action. Formal eligible-set equation, pair-selection pseudocode, complete per-library scan, and separate historical/corrected labels. Evidence location: Fig. 13B–D and the complete eligible-neighbor table; the candidate rule remains explicitly unlocked.

R2-4

Reviewer comment The claim that the cutoff “separates collinear type pairs rather than assay platforms” rests on eight single point-estimate cosine values, one per library, with CosMx PDAC as the sole counter-example demonstrating within-platform variability. This is suggestive but is not accompanied by a formal statistical test. I recommend either a supporting statistical test using the full per-neighbor-pair dataset or explicit softening of this claim’s language in the Abstract and Discussion to “consistent with” rather than a demonstrated general property.

Authors’ response and evidence

Comment. Platform-independence claim based on eight calls.

Response. We agree that the original wording overstates what eight library-level calls can establish. The new analysis treats cosine as a continuous descriptive quantity and does not fit a binary platform model using the same cosine that defines KEEP/ABSTAIN. The five designated CosMx libraries span 0.506–0.862, with both KEEP and ABSTAIN decisions on the same platform. Xenium FLEX is marked as paired with Xenium because it shares the Xenium spot design. We consequently change the claim to a qualified statement: the observed pattern is consistent with malignant-neighbor collinearity, while platform independence remains unestablished.

New evidence. B02 designated CosMx cosine spans 0.505826–0.861599 (mean 0.653568). A candidate corrected 10,000-resample library-clustered bootstrap of the complete eligible CosMx table (51 rows: CRC 15, HCC 14, NSCLC 17, PDAC 5) gives mean 0.674 [0.579,0.781], median 0.725 [0.678,0.814], SD 0.199 [0.097,0.258], and fraction ≥ 0.80 of 0.294 [0.063,0.684]. This quantifies within-CosMx library heterogeneity but is not a causal platform test; the neighbor-selection rule remains an author-lock decision, so this candidate corrected interval is not yet a replacement for the historical locked call.

Revision action. Abstract/Discussion changed to “consistent with collinearity; platform independence not established”; continuous-cosine table and confounding limitation. Evidence location: Fig. 13B–D and the library-clustered descriptive interval.

R2-5

Reviewer comment The Discussion asserts that existing composition estimators (RCTD, cell2location, etc.) “expose” no analogous reportability signal, but this is not empirically demonstrated on the same data. If feasible, a brief comparison—examining RCTD’s internal weight/confidence diagnostics or cell2location’s posterior

variance on the same CosMx BCC malignant coordinate—would strengthen the novelty claim. If out of scope, please state this explicitly as a limitation.

Authors’ response and evidence

Comment. No empirical comparison with RCTD/cell2location uncertainty.

Response. We treated this as a matched diagnostic question rather than as an invitation to rank methods by a single error number. On the common 400 spots, the malignant-coordinate RMSE was 0.154 for SPGD, 0.249 for cell2location, and 0.121 for RCTD. These values provide quantitative context, but they do not make interval width, posterior variance, and reportability interchangeable. cell2location exposes native interval widths (mean 0.0593; median 0.0659) and posterior-predictive widths (mean 0.264), while the saved RCTD output contains no matched per-spot interval or confidence column. The key result is therefore operational: in these same-input saved outputs, every comparator writes a malignant coordinate, but no comparator exposes an explicit state that withholds that coordinate while preserving the remaining simplex when the malignant program is not separable from an eligible neighbor.

This distinction strengthens the contribution rather than weakening it. The proposed gate is a decision layer around a composition estimator, not a competing posterior model and not a claim of superior RMSE. A method can have a useful interval or a low aggregate error while still forcing a point estimate for a coordinate whose reference direction is collinear. Conversely, ABSTAIN does not discard the spot: it preserves the conditional non-malignant composition and records why the malignant coordinate is absent. We have therefore replaced the blanket literature claim with a matched, bounded statement and made the comparator limitation explicit.

New evidence. Malignant RMSE: SPGD 0.154, cell2location 0.249, RCTD 0.121. cell2location interval width mean 0.0593/median 0.0659; posterior-predictive width mean 0.264; RCTD output lacks a matched per-spot interval/confidence field. The audit was performed on the same saved 400-spot coordinate and is reported as diagnostic context, not a leaderboard.

Revision action completed. The Discussion now separates uncertainty quantification from coordinate reportability and limits the novelty statement to the explicit missing-coordinate state observed in the matched outputs. Results and Fig. 14A–B identify the matched audit, its common spot set, and its method-specific fields. The new text makes the added work visible in the manuscript while preserving the central claim: SPGD-TME adds a reproducible selective-reporting contract to a composition fit.

R2-6

Reviewer comment Equation 9 renormalizes the remaining K-1 types to sum to 1 when the malignant coordinate is withheld. This implicitly assumes that the malignant fraction is missing independently of the relative proportions among the remaining types. Given that the gate is triggered by collinearity with a specific neighbor, renormalization could introduce systematic bias specifically into that neighbor’s reported coordinate on ABSTAIN spots. Please add a sentence discussing this potential bias, or, if possible, a brief quantitative check on a case where ground truth is available.

Authors’ response and evidence

Comment. ABSTAIN renormalization may bias the malignant neighbor.

Response. We agree that Eq. 9 is not merely algebraic housekeeping: it changes the estimand from an unconditional full-simplex composition to a conditional composition after withholding

the malignant coordinate. We therefore ran a forced-ABSTAIN counterfactual on the saved 400-spot estimates, leaving the historical gate decisions untouched. The retained coordinates summed to one with a maximum error of 1.1×10^{-15} . In the available truth-linked edges, conditional malignant-neighbor RMSE increased from 0.0380 to 0.186 in openST, from 0.0534 to 0.176 in realgt, and from 0.0938 to 0.189 in realgt3; the corresponding positive neighbor biases were 0.0507, 0.0249, and 0.0308, with bootstrap intervals excluding zero.

The interpretation is deliberate. We do not replace a missing malignant fraction with zero, because that would confuse “not reportable from this reference” with “biologically absent” and would create an artificial error signal. We retain the spot and report the remaining types because they still answer a conditional composition question, but we label that simplex as conditional and expose the neighbor inflation as a measurable consequence of abstention. The new result therefore turns the reviewer’s concern into part of the method’s decision contract: ABSTAIN is informative partial reporting with a known bias channel, not a silent renormalization or a discarded observation.

New evidence. Simplex closure max error 1.1×10^{-15} . Conditional neighbor RMSE: openST 0.186 [0.147,0.221], realgt 0.176 [0.146,0.204], realgt3 0.189 [0.154,0.221]; positive bias 0.0507, 0.0249, 0.0308 with intervals excluding zero.

Revision action completed. Methods Eq. 9 now states the conditional estimand and the missing-versus-zero distinction; Results reports the forced-ABSTAIN bias audit; Discussion presents it as a bounded consequence and design limitation. Fig. 14A–D and the conditional-bias table link the added analysis directly to the revised decision contract.

R2-7

Reviewer comment Locked truth (Eq. 4) is defined purely as cell-count share, treating each cell as an equal unit regardless of RNA content, cell size, or capture efficiency. This is a standard simplification but should be explicitly acknowledged as a limitation, since differential mRNA yield between large tumor cells and small immune cells could bias apparent accuracy.

Authors’ response and evidence

Comment. Ground truth is cell-count share, not transcript mass.

Response. We agree and make the estimand explicit. Equation 4 reports cell-count share, treating cells as equal units; it does not estimate transcript mass. The available saved truth does not retain the per-cell transcript-total mapping needed for an honest RNA-weighted sensitivity analysis. We therefore add the limitation that cell size, RNA yield, and capture efficiency may separate the two targets. The current evidence package records that a transcript-weighted reanalysis is not identifiable from the retained truth object; it is not presented as if it had been performed. We do not label the current cell-count result as RNA-mass accuracy.

New evidence. Saved truth contains spot-level cell-count proportions only; no cell-to-spot transcript-total mapping is available, so no RNA-weighted result is claimed.

Revision action. Equation 4 is labelled as a cell-count estimand, the cell-size/RNA-yield/capture-efficiency limitation is explicit, and Supplementary Table S5 records the truth-definition and availability boundary. No RNA-weighted result is claimed.

R2-8

Reviewer comment The stabilizing constant epsilon added inside the log for entropy calculation is never given a numerical value. Please state it.

Authors' response and evidence

Comment. epsilon is unspecified.

Response. We agree. The fixed value is $\epsilon=1 \times 10^{-12}$ in the entropy term and the associated numerical guards. This is a stability constant, not a parameter selected from the eight library calls or tuned to make the 0.80 decision work. Its role is to make zero or near-zero type probabilities finite inside the logarithm, so that a missing marker in one type does not create an undefined entropy or a platform-dependent exception. The same value is used throughout the fixed baseline and is now stated beside Eq. 1 rather than left to be inferred from rounded results.

This clarification matters for the scientific contribution because the reportability decision is intended to be a frozen, auditable contract. A reader reproducing the reference profiles can now reconstruct the entropy weights, the working signature, and the cosine values near the cutoff without guessing a hidden numerical convention. The constant is also kept separate from the cutoff: epsilon controls numerical stability of the specificity weight, whereas c^* controls the biological reportability decision.

New evidence. $\epsilon=1 \times 10^{-12}$ in the entropy term and related numerical guards; the value is listed in the fixed-constant inventory and source audit.

Revision action completed. Methods Eq. 1 now reports epsilon explicitly and explains its stability role; the exact value remains in machine-readable evidence. No estimate, threshold, or historical call changed.

R2-9

Reviewer comment The held-out generalization fraction g in [0,1] is introduced without derivation or intuition. A one-sentence explanation, even if fully derived in Fu (2026), would improve self-containedness.

Authors' response and evidence

Comment. g is introduced without derivation or intuition.

Response. We agree that “held-out improvement” was too compressed. In the fixed baseline, spots are permuted with a fixed seed and split into halves A and B. The platform correction is estimated on A and evaluated on B, the direction is reversed (BtoA), the two relative held-out Poisson-deviance reductions are averaged, and the result is clipped to [0,1]. Thus g is the transferable fraction of a platform correction: near zero means the correction does not generalize beyond the half that estimated it, while a larger value permits more of that correction into the working signature because it improves held-out fit.

The cross-half construction is also a safeguard against a hidden second training stage. g uses the fixed baseline objective and no outcome labels, does not tune the malignant-neighbor cutoff, and is not selected from the reported KEEP/ABSTAIN calls. It makes the platform adjustment self-limiting: an in-sample improvement is not automatically treated as a library-wide correction. The revised wording therefore gives readers enough intuition to understand why g enters Eq. 2 while preserving the distinction between baseline provenance and the new reportability operator.

New evidence. Two-way permuted half split; relative held-out Poisson-deviance reductions averaged and clipped to [0,1].

Revision action completed. Methods Eq. 2 now states the split direction, averaging, clipping, and transfer interpretation; the baseline-provenance record carries the same definition. This makes the inherited computational stage self-contained without claiming that the present paper derives a new platform model.

R2-10

Reviewer comment Wall-clock timings are reported to millisecond precision but appear to be single-run measurements on a single unspecified machine. Please report CPU/hardware specification and mean $\pm$ SD over repeated runs.

Authors' response and evidence

Comment. Single-run millisecond timings are not reproducible.

Response. We repeated the existing 400-spot openST build after a warm-up pass and obtained 22.565 ± 2.861 s across five runs (CV 12.68%; range 19.026–25.588 s). The in-memory refusal operation required 0.455 s for 1,000 iterations. The measurement was made on an Intel Core Ultra 9 275HX with 24 logical CPUs and 62 GiB RAM, using Python 3.13.7 and no accelerator. We report seconds, mean $\pm$ SD, CV, range, warm-up policy, and the machine-specific scope rather than presenting millisecond precision as an intrinsic property of the method.

The repeated measurement also clarifies what the added operators cost. The end-to-end value includes the fixed reference construction and composition fit, whereas the refusal operation is a small in-memory decision step. Thus the proposed selective-reporting layer is computationally inexpensive relative to the baseline fit, while the full runtime remains an input-size- and hardware-dependent engineering quantity. This separation is useful for readers deciding whether to apply the gate to a larger library: the scientific decision contract does not require a second expensive model fit.

New evidence. Fixed-baseline openST 400-spot run: 22.565 ± 2.861 s, CV 12.68%, range 19.026–25.588 s; refusal loop: 0.455 s for 1,000 applications.

Revision action completed. Methods Timing, Table 2, and the timing evidence now report hardware, Python version, warm-up, repeated mean $\pm$ SD, CV, and second-scale precision. The manuscript labels the benchmark as machine-specific while using it to quantify the overhead of the new operator.

R2-11

Reviewer comment The builder floor of 50 reference cells per type is stated but not justified. A brief sensitivity check, for example 25 versus 50 versus 100 cells, would strengthen confidence that downstream results are not sensitive to this choice.

Authors' response and evidence

Comment. Floor of 50 reference cells is unjustified.

Response. We agree that support retention alone cannot prove downstream invariance. We therefore ran an isolated downstream extension using the same donor construction and fixed baseline. The extension covers all three directed edges at floors 25, 50, 75, and 100. Floors 50, 75, and 100 produced identical estimates on each edge because the retained reference support did not change. Moving from floor 25 to 50 changed malignant RMSE from 0.159 to 0.163 for A-from-B, from 0.102 to 0.103 for B-from-A, and from 0.138 to 0.122 for D-from-C; corresponding PCC values were 0.853 to 0.850, 0.976 to 0.975, and 0.941 to 0.945. These edge-dependent results quantify local sensitivity for this donor construction without claiming universal floor invariance.

New evidence. Twelve downstream fits completed: three directed edges at each of floors 25, 50, 75, and 100. The malignant-coordinate RMSE sequences at floors 25/50/75/100 are A-from-B 0.159/0.163/0.163/0.163, B-from-A 0.102/0.103/0.103/0.103, and D-from-C 0.138/0.122/0.122/0.122; the corresponding PCC sequences are 0.853/0.850/0.850/0.850,

0.976/0.975/0.975/0.975, and 0.941/0.945/0.945/0.945. No edge/floor combination remains pending in this bounded extension.

Revision action. Add the floor rationale and distinguish support retention from the bounded downstream sensitivity result. Evidence location: Supplementary Table S7 and the completed F14C support panel.

R2-12

Reviewer comment Numerical precision throughout the main text and Abstract is reported to 5-6 significant figures. For a threshold-based binary decision this reads as excessive precision in prose; consider 3-4 significant figures in the main text and Abstract, reserving full precision for supplementary tables.

Authors' response and evidence

Comment. Five-to-six significant figures in prose.

Response. We agree and applied a manuscript-wide precision policy. The binary reportability decision is governed by the relation to $c^*=0.80$, so prose reports the decision and a readable three- or four-significant-figure cosine; it does not imply that the sixth decimal has biological meaning. Continuous audit metrics are treated differently: exact values are retained where they document a threshold margin, a bootstrap calculation, a simplex-closure check, or an independently reproducible table entry.

This separates communication precision from computational precision. Rounding the Abstract, Results, and figure captions improves readability without changing any stored estimate, threshold comparison, or KEEP/ABSTAIN call. The exact values remain in the supplementary/machine-readable evidence so that a reader can reproduce the sensitivity grid and distinguish a comfortably separated call from a cutoff-adjacent candidate scan. We also retained the 1.1×10^{-15} closure result in the audit record because it is a numerical invariant, not a biological effect size.

New evidence. Narrative values are now rounded to three or four significant figures; exact cosine, bootstrap, timing, and closure values remain in the audit tables and integrity records.

Revision action completed. Abstract, main Results, captions, and response text received the precision pass. The change improves interpretability and presentation while leaving the scientific estimates and decisions unchanged.

R2-13

Reviewer comment The 95% bootstrap interval ($B = 1000$) is computed on a 400-spot subsample even where full-n data are available. Please justify the choice of 400 rather than the full sample, or note this explicitly as a design choice.

Authors' response and evidence

Comment. Why bootstrap 400 when full n is available?

Response. We agree that the original manuscript did not distinguish a bounded audit sample from a full-library confirmation. The 400-spot design is a fixed computational budget for the interpolation, bootstrap, and cross-substrate comparison: it keeps the resampling calculation inspectable and makes the three dose-response substrates comparable under the same spot count. It is not presented as a replacement for the full library. Where a saved full-n estimate exists, we now use it as a direct sensitivity check rather than leaving the reader to assume that the bounded interval is representative.

On realgt3 with B=1000, malignant-coordinate RMSE was 0.154 [0.135, 0.175] on the 400-spot audit and 0.152 [0.148, 0.156] on all 5,686 spots. The close point estimates, together with the narrower full-n interval, show that the 400-spot result is a bounded reproducibility instrument rather than a selective choice made to obtain the reported value. The bootstrap resamples saved estimates; it does not refit the model, and nearby spatial spots may be correlated. We now state both facts so the interval is interpreted as a saved-estimate uncertainty summary, not as an independent-cell confidence interval.

New evidence. realgt3 malignant RMSE: 400 spots 0.154 [0.135, 0.175]; full n=5,686, 0.152 [0.148, 0.156]. The full-n confirmation changes neither the direction of the result nor any KEEP/ABSTAIN call.

Revision action completed. Methods and Results now explain the fixed 400-spot budget, identify the full-n sensitivity, and state the saved-estimate/spatial-dependence boundary. Supplementary Table S8 and Fig. 14 include the paired 400-versus-full-n values. The added analysis strengthens the reproducibility claim without presenting a computational convenience as a biological sampling guarantee.

R2-14

Reviewer comment The per-library fraction of neighbor pairs exceeding the cutoff is one of the more persuasive pieces of evidence for the paper's central claim, yet it currently appears only within a figure caption. Consider promoting this to a main-text table for visibility.

Authors' response and evidence

Comment. Neighbor-pair fractions are mainly in a caption.

Response. We agree that a reader should be able to inspect the denominator and numerator without extracting them from a caption. The revised Results paragraph now states the historical stored-scan fractions directly: 2/17 for NSCLC, 0/8 for the historical CRC export, 8/15 for HCC, and 6/6 for PDAC. We also expose the complete eligible-neighbor table used in the revision audit, including the recovered fifteen CRC eligible neighbors. This presentation makes the evidence visible while preserving the distinction between an historical stored scan and a candidate complete eligible scan.

The denominator distinction is scientifically important. Treating the historical CRC 8-row export as if it were a complete 15-neighbor scan would create a false platform comparison. We therefore keep the historical fractions as descriptive records, label their provenance in the table and figure caption, and report the candidate complete-scan maxima separately until one neighbor-selection rule is author-locked. This is a stronger reproducibility response than silently replacing the original fractions after seeing the new scan.

New evidence. Historical stored-scan fractions are 2/17, 0/8, 8/15, and 6/6; the candidate eligible table contains 15 CRC, 17 NSCLC, 14 HCC, and 5 PDAC rows. The complete eligible table exposes each pair, its cosine, eligibility, and cutoff status.

Revision action completed. The historical fractions are promoted to the main Results text and Fig. 13; the complete library-by-neighbor table is included in the supplementary evidence package. Corrected denominator-sensitive fractions remain explicitly marked as candidate until the selection rule is locked. The remaining action is an operating-rule decision, not a missing result.

R2-15

Reviewer comment At 7,012 words, 12 figures, and 3 tables, this is substantial for a Methods Article. Please confirm that this conforms to the journal’s format limits, or consider relocating descriptive rather than evaluative figures to supplementary material.

Authors’ response and evidence

Comment. Length/scope and article-type limits.

Response. We agree that the editorial package should distinguish scientific completeness from the final float allocation. The V2 review build is 7,061 words, 14 figures, and 3 tables (17 display objects). It is deliberately expansive so that the new cutoff, neighbor-scan, comparator, conditional-bias, floor, timing, and bootstrap analyses can be inspected together during review. The added evidence is carried primarily by panels inside the existing figure architecture, rather than by a sequence of disconnected new figures.

The review build therefore serves two purposes: it gives the reviewer a complete view of the additional work, and it provides a stable source from which a journal-format package can be assembled. The final float pass will preserve the evaluative panels for the two operators and move descriptive geography or audit-only tables to the supplement if the journal counts tables and figures as a combined limit. This is a packaging decision, not a reason to remove the quantitative results from the review record. The historical V0/V1 builds remain preserved, so the final allocation can be changed without losing the expanded evidence or the original presentation.

New evidence. The review PDF now states 7,061 words and carries 14 figures plus 3 tables; F13/F14 and the supplementary audit tables contain the added quantitative results. The current build is explicitly labelled a review build rather than silently claiming that 17 display objects satisfy a 15-object final cap.

Revision action completed. The manuscript and response now state the actual review-build counts, keep the central operator panels together, and define a main-text/supplement partition rule. The remaining editorial step is to apply the journal’s final float-count convention after the author-lock of R2-3; no scientific result depends on that packaging step.

R2-16

Reviewer comment Please confirm that the NanoString data license governing the Zenodo redistribution permits the described derived-reference-matrix use and digest-based identification, and that this satisfies the journal’s data-sharing policy.

Authors’ response and evidence

Comment. NanoString license and derived-reference/data-sharing permission.

Response. We examined the public Zenodo record and its attached license artifact. The record identifies a file named NanoString Data License Agreement.pdf, gives its MD5 identifier, and explicitly directs users to that agreement for the CosMx liver and lung terms. The record also states that the CosMx datasets originated from NanoString and that reuse is subject to the attached terms. This establishes the provenance trail and the existence of a governing agreement; it does not, by itself, prove that our derived reference matrices may be redistributed.

We therefore separate three permissions that are often conflated: (i) using the source data to perform the analysis, (ii) distributing derived reference or plot-data objects, and (iii) distributing integrity metadata that identifies an artifact. Our V2 Data Availability statement authorizes none of the second or third categories by inference. It makes the derived-reference release conditional on the agreement text and the journal’s policy, while the public archive can still

provide the software, locks, renderers, tests, and non-restricted analysis outputs. This wording is intentionally precise: it protects the reproducibility claim from being stronger than the underlying data license.

New evidence. The Zenodo record is open, dated 27 June 2025, and lists the attached NanoString agreement as a 179.5 kB PDF with MD5 25c95f9721b6e5a34b1814ff1100c1d9. The record description says “See NanoString_Data_License_Agreement.pdf for the license terms” for both CosMx liver and CosMx lung. Clause-level permission for derived-reference redistribution remains unasserted until the agreement body is directly read through an authorized route.

Revision action completed. Data Availability and portal metadata now identify the agreement as the controlling permission source, distinguish software/analysis artifacts from licensed raw or derived data, and avoid any affirmative redistribution claim. The remaining task is a legal-text inspection or author-supplied permission confirmation, not another model experiment.

R2-17

Reviewer comment Citation Fu, 2026 carries a publication year later than several other citations and later than the manuscript’s own submission-date context. If this is in-preparation, in-review, or software-archive-only, please label its status explicitly rather than citing it as a standard dated reference.

Authors’ response and evidence

Comment. Fu (2026) year/status appears later than the submission context.

Response. We agree that a software archive should not be formatted as though it were a peer-reviewed article. The revised reference and first citation identify Fu (2026) as a public software archive for the released baseline implementation, with its DOI and access date. The wording does not imply acceptance, peer review, or an independent journal publication. This also clarifies the relationship between the archive and SPGD-TME: the archive documents fixed baseline provenance, while the present manuscript contributes and evaluates the locked reportability and donor operators.

This status label is not a retreat from the method’s contribution. It prevents a bibliographic date from carrying an unintended evidentiary meaning and lets the reviewer evaluate the operator-inclusive analyses on their own terms. The manuscript now uses the software record for provenance, not as a substitute for the evidence generated in this article.

New evidence. The local reference record supports the description “public software archive for the released baseline”; no separate peer-reviewed article is asserted by this citation.

Revision action completed. The reference entry, first citation, Methods provenance paragraph, response letter, and portal text use the explicit archive status. The DOI and access date remain available for retrieval.

R2-18

Reviewer comment I spot-verified several key figures across the Abstract, Results, and Discussion and found them numerically consistent. This is good practice and worth noting positively.

Authors’ response and evidence

Comment. Key numbers are internally consistent.

Response. We thank the reviewer for identifying this strength. We treated the positive spot-check as a release criterion rather than as a one-time compliment. After the V2 Methods, Results, Discussion, F13/F14 panels, and response artifacts were regenerated, we reran the cross-section number audit. The historical values remain unchanged: the wound-axis tumor fraction is 0.8425 to 0.4665, B-from-A RMSE is 0.0659, and B-from-A spot-level PCC is 0.9528. The new values are tracked separately so that the revision audit cannot silently overwrite the historical result.

The check covers both directions of consistency: narrative-to-table and table-to-plotdata. It also checks the decision semantics, not just decimal equality: the same cosine remains below or above the frozen $c^*=0.80$ cutoff in every cited location, and the ABSTAIN coordinate remains represented as missing rather than zero. This makes the number audit relevant to the paper's central contribution, because a reportability method is only useful if its decision, estimate, caption, and response letter agree.

New evidence. Cross-section audit after the final V2 response regeneration: 0.8425 to 0.4665, RMSE 0.0659, PCC 0.9528; the exact machine-readable values and closure checks remain in the capture evidence. The focused contract suite passed 31 tests, and the capture/mirror integrity inventories verify.

Revision action completed. The consistency audit is now described as a release control in the Lighthouse and verification ledger. The final PDF, response PDFs, portal text, manuscript source, and integrity inventory were regenerated after the expanded R2-13–R2-18 sections, so the positive reviewer observation has been converted into an explicit quality-control step.

Scope of the revised claim

The revised manuscript no longer presents platform independence as demonstrated. It presents the observed within-CosMx spread as consistent with malignant-neighbor collinearity, explicitly reports the pair-selection ambiguity, treats ABSTAIN renormalization as a conditional estimand, and labels the cell-count truth, saved-estimate bootstrap, method-specific uncertainty fields, and unresolved license terms with their appropriate limitations.